Preme Malaviphusit

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EDUCATION

New York University, Tandon School of Engineering | Brooklyn, NY — May 2028

Bachelor of Science in Electrical and Computer Engineering (Dual Major) | GPA: 3.73

- Relevant Coursework: PCB Design, Embedded Systems, Computer Architecture, VLSI (Graduate), Signals and Systems, Digital Logic, Electronics, Object-Oriented Programming, Data Structures and Algorithms

EXPERIENCE

Teaching Assistant, Head of Prototyping – *NYU General Engineering* | Brooklyn, NY — Aug 2025 - Present

- Managed Prototyping Lab for 400+ students and trained TAs in embedded systems, soldering, CAD, and 3D printing
- Debugged 50+ Arduino, ESP32, and Raspberry Pi prototypes, fixing shorts, cold joints, and firmware bugs
- Led weekly hands-on labs for 16 students covering circuit design, firmware development, CAD, and data analysis
- Mentored 4 project teams through full hardware-software integration, resulting in 100% on-time prototype delivery

Computer Vision Researcher – *Flexible AI-Enabled Mechatronic Systems Lab* | Brooklyn, NY — Sep - Dec 2025

- Developed a kidney stone detection pipeline using TensorFlow and PyTorch for real-time endoscopic imaging
- Generated 3D volumetric heat maps in 3D Slicer to support laser ablation planning

Electrical Engineering Intern – *Metropolitan Electricity Authority* | Bangkok, Thailand — Jun - Jul 2023

- Produced 10+ AutoCAD electrical drawings and load analyses for the Thai Parliament substation
- Inspected 5+ electrical systems across solar, EV, and hospital infrastructure for performance and compliance

TECHNICAL PROJECTS

Payload Electronics & RF Telemetry - *NYU Rogue Aerospace* | Altium Designer, C++, XBee — Jan 2026 - Present

- Developed firmware and electronics for autonomous water-ejection payload, advancing from prototype to flight-ready 4-layer PCB; deployed successfully at apogee (~1,900 ft) on NYU's first hybrid rocket flight
- Integrated XBee RF telemetry to downlink tank pressure, altitude, and GPS coordinates to ground station

RTL Verification and Optimization Agent - *Hack the Chip (NYU x Annapurna Labs)* | Python, Verilog — Oct 2026

- Built LLM agent (Qwen3-8B on AWS Trainium datacenter chips) that writes, debugs, and optimizes Verilog
- Developed self-correcting Icarus Verilog loop; auto-generated testbenches caught 93% of mutation-injected bugs
- Integrated Yosys-guided optimizer, shrinking designs up to 38% (61 → 38 cells) with full re-verification

SharkFin Rev 1.0 Miniaturized Avionics Board - *NYU ARC Robotics* | Altium Designer, STM32 — Jan - Sep 2026

- Brought up 4-layer STM32H743 PCB for guided projectile stabilization with IMU, OV5640 camera, and SD logging
- Implemented dual-input USB-C/LiPo mux with reverse-polarity protection and multi-rail LDO (3.3V/2.8V/1.3V)
- Routed 0.8mm pitch BGA fanout and high-speed DCMI interface within flight-form-factor constraints

UAV Flight Controller Board | Altium Designer — Apr - May 2026

- Designed 4-layer STM32F405 flight controller with SPI IMU, power monitoring, and 4-channel PWM ESC control
- Architected 6S LiPo to 5V@5A buck with MOSFET reverse-polarity, TVS clamping, and USB/battery power mux
- Routed 90Ω USB differential pairs and 5A power paths with solid GND planes, via stitching, and JLCPCB DFM rules

LEADERSHIP / EXTRACURRICULAR

Secretary, NYU Society of Asian Scientists and Engineers — May 2026 - Present

Vice President, NYU Society of Women Engineers | Outstanding Event Management Award — Oct 2025 - Present

President, NYU Thai Student Association | Represented NYU at 6,000+ attendee university fair — Jul 2025 - Present

SKILLS

Altium Designer, Cadence Virtuoso, LTSpice, Vivado, Multimeter, Oscilloscope, Logic Analyzer, Soldering, SolidWorks, STM32, SPI/I2C/UART/CAN, Python, C/C++, SystemVerilog, MATLAB, Linux, Git